CODE BASICS

DEEP LEARNING TAKEAWAYS

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Chapter:

Neural Networks: Fundamentals

What is Neuron?

1. Neurons are the basic building blocks of neural networks in deep learning.
2. Each neuron receives input, processes it, and passes it to the next layer.
3. Neurons apply a weight to the input and use an activation function to determine the output.
4. Logistic regression can be thought of as a simplest form of a neural network with a single neuron.
5. A neuron in deep learning mimics the biological neuron in the brain.

Perceptron and Multilayer Perceptron

1. The perceptron is the simplest neural network model that classifies linearly separable data
2. The multilayer perceptron (MLP) extends the perceptron with hidden layers and non-linear activation functions.

Neural Network Intuition: Insurance Prediction

1. Neural networks have input, output and hidden layers
2. For the insurance example, features such as Age and Education helps fire Awareness neuron in a hidden layer. The other features such as Income and Savings help fire Affordability neuron in a hidden layer
3. If a person will buy insurance or not will be determined based on Awareness and Affordability
4. At each stage starting from input to output, a neural network is extracting patterns from the data so that it can make a final decision using those patterns

The Purpose of Activation Function

1. Real world problems are often non-linear in nature and activation function helps you introduce non-linearity in a neural network

2. With activation functions, neurons either “fire” and “do not fire”. This can be thought of as individual detectives who are given a specific task and after investigation they give their findings to a judge (which is the neuron in the next layer).

3. In case of insurance prediction example, in the hidden layer we have two detectives, one responsible for figuring out awareness and the other one for affordability. They give their conclusion that person’s awareness is 0 or 1 (if using step activation function), or it is a number between 0 to 1, say 0.7 (when using sigmoid activation function)

Activation Functions: Sigmoid, ReLU, Tanh, SoftMax

1. Sigmoid, Softmax, tanh, ReLU are the most commonly used activation functions
2. Sigmoid is primarily used in output layer for binary classification problems (e.g. will person buy insurance, is the transaction fraud)
3. Softmax is primarily used in output layer for multi-class classification problems (e.g. handwritten digits classification, clothes classification)
4. tanh is similar to Sigmoid, the only difference is that the output range is -1 to 1 (for sigmoid the range is 0 to 1)
5. ReLU is a default choice for neurons in hidden layers as it is fast to calculate and also doesn't suffer much from a vanishing gradients problems
6. Leaky ReLU is a variant of ReLU that helps with dead neurons problem